# Zain ul Abideen

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**Research interests:** post-training and reinforcement learning for language and vision-language models — reward design for structured and verifiable outputs, on-policy distillation, and evaluation of clinical reasoning and safety behaviour in medical LLMs.

## EDUCATION

**BS Software Engineering** — National University of Sciences and Technology (NUST) 2021 – 2025

- CGPA 3.55 / 4.00. Islamabad, Pakistan.

## EXPERIENCE

**Machine Learning Engineer** — John Snow Labs Dec 2024 – Present | Remote

- Lead post-training for the JSL-Med LLM family (4B, 8B, 30B, 70B) — SFT, on-policy distillation, preference optimization, and RLVR — building the data curation, reward design, and training pipelines behind each release. The objective across the family is improved medical reasoning, clinical coherence, and calculation, with safety behaviour held to the same bar: correct handling of protected health information and confidential clinical data. JSL models rank No. 1 on the Open Medical LLM Leaderboard.
- Built the vision-language extraction stack for structured clinical data from handwritten medical documents. SFT followed by field-aware asymmetric OPD, plus constrained decoding against 12k medical entities. Medical LLM Small reached within a point of the teacher at a third the parameter count.
- Own medical evaluation end to end — clinical decision support, clinical calculation from patient notes, medical error detection, race-bias in clinical recommendations, plus oncology, clinical entity extraction, and multilingual medical benchmarks; benchmarking JSL-Med against frontier models.
- Compressed a 125B sparse mixture-of-experts model to single-H100 size using Cerebras REAP expert pruning, then trained the pruned model back to 95% of baseline medical accuracy — at roughly twice the token cost per answer.

**Visiting Research Intern (Mitacs)** — University of Alberta Jun – Sep 2024 | Edmonton, Canada

- Worked on a vision-language pipeline for automated structural analysis, taking images of structural systems as input and generating executable OpenSeesPy finite-element code as output. Supervised by Prof. Qipei (Gavin) Mei.

**Machine Learning Engineer (Part-time)** — Danucore Jun – Nov 2024 | London, UK — Remote

- Built a self-improving multi-agent pipeline that takes a single user prompt and runs to a finished result without a human in the loop: agents plan the task, execute it, critique their own output through an LLM-as-a-Judge loop, and re-run until the result converges.
- Combined RAG for grounding with a Mixture-of-Agents layer for reasoning, spanning text and image generation across open-source foundation models — Llama-3.1-70B, Phi-3-Medium-128K-Instruct, LLaVA-Next-8B, SDXL-Lightning — served on 30+ GPUs across multi-node infrastructure.

**Machine Learning Engineer (Part-time)** — QueryloopAI Nov 2023 – Jun 2024 | SF Bay Area — Remote

- Pre-trained a 500M SLM from scratch on a carefully curated high-quality 15B-token synthetic dataset. Created the entire training and evaluation pipeline along with managing training on 8×A100s.
- Experimented with and contributed to various LLM open-source projects including Mixture of Experts, Hybrid MoE, BitNet, Mixture of Depths, LaserQLoRA, and DASER.
- Headed the LLM.pth open-source project, a library providing implementations of various autoregressive models, research papers, and techniques in PyTorch.

## SELECTED OPEN-SOURCE PROJECTS

- **llm.pth** — wrapper-free PyTorch implementations of autoregressive architectures (Llama, Mixtral, Gemma, DeepSeek), research methods (CoPE, YaRN, MoD, MoME, MLA), and post-training techniques (SFT, DPO, KTO, IPO).
- **llama3.cuda** — Llama 3.1 implemented in pure C/CUDA, with hand-written SwiGLU, RoPE, RMSNorm, and grouped-query attention kernels.
- **prime-lab-trainer** — automated pipeline that turns any HuggingFace dataset into a live GRPO training run on Prime Intellect, handling environment construction, reward-function validation, and submission.
- **AutoSynth** — synthetic data generation implementing Self-Instruct, Magpie, Agent Instruct, Arena Learning, Genstruct, and self-curation.
- **LightAgents** — dependency-free agents library with RAG, function calling, JSON mode, telemetry, and multi-layer memory.

## OPEN-SOURCE MODELS & DATASETS

- 31 models and 28 datasets released on Hugging Face. Notable: **Bitnet-Llama-70M** (1.58-bit quantization-aware training), a controlled comparison of **LASER / DoRA / DASER / LoRA** PEFT methods on a common base, **MonarchCoder-MoE-2x7B**, and Phi-3-mini aligned with CPO-SimPO.
- Curated and published training datasets including fineweb-edu-100k, FineTome-Alpaca-30k, LongCite-English-Filtered-13K, and medical SFT sets for clinical calculation.

## TECHNICAL WRITING

- 42 articles on LLM architecture and training, 1.3k followers on Medium. Selected: *MoE vs Dense vs Hybrid LLM Architectures* (trained 600M models of each for a controlled comparison); *MHA vs MQA vs GQA vs MLA*; *Linear RoPE vs NTK vs YaRN vs CoPE*; *Coding DeepSeek-V2 from Scratch in PyTorch*; *Multi-GPU Training of 70B LLMs with DeepSpeed and FSDP+QLoRA*; *Best LLM Inference Engine? TensorRT vs vLLM vs LMDeploy vs MLC-LLM*.

## TECHNICAL SKILLS

- **Training & post-training:** pretraining from scratch, SFT, DPO / KTO / IPO / CPO-SimPO, GRPO, RLVR, RLHF, on-policy distillation, LoRA / QLoRA / DoRA, constrained decoding, reward design.
- **Systems:** PyTorch, CUDA / C, DeepSpeed, FSDP, TRL, vLLM, TensorRT-LLM, LMDeploy, multi-node GPU training (8×A100, H100).
- **Tooling:** Hugging Face stack, Weights & Biases, LangGraph, Python, Docker, Git.